

Review of Load Forecasting Methods for Electric Vehicle Charging Station

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Abstract—As a major tool for reducing carbon emissions, the ownership of electric vehicles is increasing year by year. When EV numbers reach a certain level, it is equivalent to a huge load randomly connected to the network, bringing new challenges to the daily monitoring and management of EV charging stations. Accurate load prediction methods for EV charging stations with high reliability are the basis for ensuring safe operation of charging stations and supporting the planning decisions of new construction and expansion of charging infrastructure. In this paper, the charging station load prediction methods are divided into three categories: charging station angle prediction, unsupervised learning charging load prediction and supervised learning charging load prediction, and the representative studies on load prediction for EV charging stations in these categories are reviewed respectively.

Keywords—electric vehicle charging stations, load forecast, machine learning, SVM, deep learning, GAN

I. INTRODUCTION

With the shortage of oil resources and global warming caused by the yearly rise in carbon emissions, it has become a consensus to use renewable energy instead of non-renewable fossil energy [1]. Electric vehicles are powered by energy storage batteries, considerably reducing the consumption of non-renewable energy sources such as oil [2], and do not pollute the environment throughout their life cycle. As the acceptance of electric vehicles increases and the electric vehicle industry continues to grow, the larger battery capacity of electric vehicles and the random selection behavior of users make it more challenging to make accurate predictions of

charging station loads.

Accurate load forecasting is the basis for the development of intelligent control of electric vehicle charging-related facilities [3], while accurate load forecasting is also responsible for the operation as well as the daily maintenance of the power system [4]. Accurate power load forecasting will greatly save maintenance costs during daily operation and is an important measure to improve the safe operation of charging stations, and also provides an important decision basis for the safety monitoring of smart grids, as well as a strong support for new charging infrastructure investment and charging station capacity expansion and planning decisions [5]. Unfortunately, the accuracy of power load forecasting can be affected by many factors such as climatic conditions, national policies, economic development, and human social activities, resulting in a degree of accuracy that does not meet the needs of the actual situation [6].

There are no fixed criteria for the time scale of load forecasting. Most of the authors tend to divide load forecasts into three categories in terms of time scales according to the time of forecast demand [3,7-9]: short-term forecasts, medium-term forecasts and long-term forecasts. Some authors also prefer a more refined four-category division [4,10-12]: ultra-short-term forecasts, short-term forecasts, medium-term forecasts, and long-term forecasts, as shown in Table 1.

Currently for accurate forecasting of charging station loads a large number of researchers have moved from traditional forecasting methods based on mathematical forecasting models, including convergent extrapolation, elasticity coefficient methods, time series methods, linear regression methods, etc., to machine learning type algorithms extending from artificial intelligence, including decision trees, neural networks, integrated learning, deep learning, etc.

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In this paper, a review and synthesis of published papers related to the field of EV charging station load forecasting is provided. In Section 2, literature related to the simulation of EV-related features for load forecasting is presented. Section 3 presents the literature on unsupervised learning based charging station load forecasting. The literature related to supervised learning based charging station load forecasting is presented in Section IV. Section V provides a summary of the reviewed literature.

TABLE I. ELECTRIC VEHICLE CHARGING STATION LOAD FORECASTING TIME SCALE CLASSIFICATION AND USAGE SCENARIOS

<i>Prediction Method</i>	<i>Time Scale</i>	<i>Usage Scenarios</i>
Ultra/very short-term load forecasting (VSTLF)	Within one hour, even for real-time control	Real-time monitoring of power systems.
Short-term load forecasting (STLF)	One hour to one week	Daily operation and maintenance of power service related companies.
Medium-term load forecasting (MTLF)	One week to one year	Maintenance and repair of power-related facilities and fuel procurement plans, as well as energy trading and profit evaluation of power service companies.
Long-term load forecasting (LTLF)	The next year to 20 years	Construction of new cities and development planning of generating units, transmission systems and distribution systems.

II. A BRIEF REVIEW OF LOAD FORECASTING BASED ON AN ELECTRIC VEHICLE PERSPECTIVE

As an emerging field, many researches focus on the characteristics of charging station load in time and space, and load forecasting by building a model to simulate the variation pattern of charging station load. Xiaolin Ge learns the EVCL (Electric Vehicle Charging Load) data of several EVs and charging stations with different characteristics through IRF, and models the data obtained after learning using a parallel model to optimize the computation process, and constructs a model with higher prediction accuracy than the original model, and shows the distribution of charging load in a certain area more clearly [13]. Kejun Qian developed an integrated analysis method for load prediction using real-time tariff variation and EVCL variation data [14]. Nima Ghasnezhad used real EVCL data to extract charging behavior probabilities and After that, the charging behavior is simulated by inputting into a fuzzy decision system and the developed model proves practical in experiments [15]. Shaoyin Chi extracts charging characteristics from EV model and speed characteristics of EV from traffic network model, after that the EVCL is predicted by simulating the movement characteristics of EV within a single day [16]. Md Shariful Islam extracts the charging characteristics from statistical data features from previous data of EVCL, and afterwards used these features to simulate the SOC distribution of EVs and finally predicted EVCL by SOC

before day [17]. Hongcai Zhang quantified the electric vehicles with parking demand by PGRM method, and discussed the PEV charging demand according to the different parking behaviors possessed by different land use neighborhoods and their charging facilities respectively, and finally proposed a simulation procedure based on Monte Carlo algorithm idea to simulate the parking, driving and charging behaviors of PEVs, and then made load prediction based on the results obtained from the simulation [18]. Nikita Korolko simulated the working state of fast chargers by extracting the behavioral empirical data of EV users and later simulated the EVCL of users using the time series features obtained from this simulator [19].

III. A BRIEF REVIEW OF UNSUPERVISED LEARNING-BASED LOAD FORECASTING MODELS

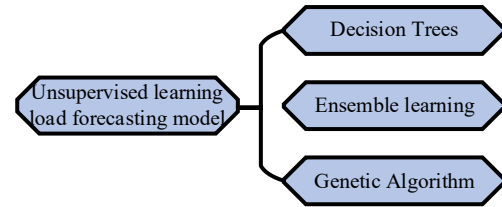


Fig. 1. Classification of unsupervised load forecasting models

A. Decision Trees

The decision tree based charging station load forecasting method with its smaller arithmetic power requirement and unsupervised convenience has been used a lot at the beginning of the development of machine learning. Yiqi Lu analyzed the feature data of EVCL from many perspectives by combining random forest algorithm with categorical regression tree algorithm, and completed a lot of learning based on the analysis results, and finally obtained good prediction results [20]. Xiaoqun Liao constructed a similar daily load forecasting model based on the idea of XGBoost, which can control the complexity and overfitting and obtain good forecasting results, by basing on the influence of weather and day type on the load and the pattern that this influence may occur [21]. Raza Abid Abbasi converted daily load data of Australian energy market operators into weekly data time series and used XGBoost to extract features from the data. XGBoost was then used again to predict lagged electrical loads in isolated times, and the method outperformed other schemes in terms of average percentage error metrics [22]. Meiqin Mao used a parallelized random forest weight algorithm to model the SC (dispatchable capacity) of a population EVCL, and the model showed advantages in speed and accuracy in comparison experiments with SVM [23].

B. Ensemble Learning

Ensemble learning algorithms substantially replace single-model algorithms with their good prediction accuracy. Xingshuai Huang used a linear regression (LR) algorithm for learning the weights of each base learner and later combined different three base learners to obtain better prediction accuracy [24]. Songpu Ai separately predicted the time of occurrence of day-ahead household EVCL and "Kai Liu developed an EVCL prediction method called ESG using a stacking algorithm that integrates decision trees, random

forests, and K-nearest neighbors, and achieved better classification performance than a single algorithm in an experimental case, and obtained higher robustness than a single model in an experimental case [26].

C. Genetic Algorithm

The genetic algorithm also achieved good improvement by combining with other algorithms. Shaobing Yang used the ant colony (AC)-based parameter identification algorithm to discrete the charging flow of EVCL into a sequence to simulate the charging state of EVCL and obtained good charging load prediction [27]. Yunyan Li used ecotone immunity to limit the lion swarm algorithm to limit the over-replication phenomenon in the algorithm, and resets the weights and thresholds in the convolutional neural network according to this phenomenon, and this short-term EVCL hybrid prediction model has higher accuracy with higher generalization ability in the experimental case [28]. Ping-FengPai optimized the parameters of SVM by simulated annealing algorithm to constitute the EVCL prediction model named SVMSA. This algorithm gives better results in the analysis of numerical cases compared to the autoregressive integrated moving average algorithm and the traditional regression neural network algorithm [29].

IV. A BRIEF REVIEW OF SUPERVISED LEARNING-BASED LOAD FORECASTING MODELS

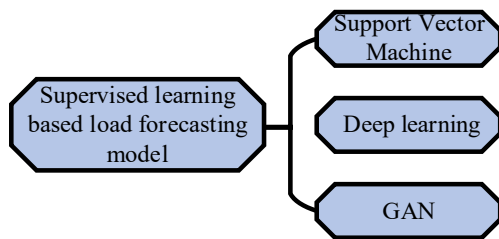


Fig. 2. Classification of supervised load forecasting models

A. Support Vector Machine

SVM (Support Vector Machine) is a class of generalized linear classifier, which is used extensively in the field of charging station load prediction. Qiming Sun used SVR to perform spurious data before training model building after considering the historical data of charging load, the number of electric vehicles, the number of charging posts in working condition, weather information, holiday category and other information correction, effectively improving the accuracy of charging load prediction [30]. Zhang Xing extracted the features of electric bus charging station load by fuzzy clustering algorithm, and iterated and optimized the parameters of LSSVM by WPA algorithm. Finally, a hybrid EVCL prediction model is constructed based on the experimental results, and the constructed model is experimentally verified to have a better prediction effect than the single model.[31]. Xiaoqun Liao used particle The parameters of the SVM were reselected using particle swarm optimization (PSO), and the PSO-SVM showed sufficient prediction accuracy under the normalized root mean square error (NRMS) criterion in numerical experiments [32]. model overcomes the drawbacks of traditional models and achieves high prediction performance [33].

B. Deep Learning

With the development of deep learning, more researchers are also inclined towards deep learning based EVCL prediction algorithms. Tim Unterluggauer used long and short-term memory (LSTM) modeling to predict the overall charging load at each charging point over a fifteen-minute range by using real data from public locations such as shopping malls and residences, and obtained good prediction results [34]. Miguel López focuses on an online moderated EVCL forecasting model based on a combination of autoregressive models and neural networks, which is currently used in the Spanish transportation system, to forecast hourly and real-time electricity loads in 18 regions of Spain for the current day and future periods[35]. Juncheng Zhu compared four deep learning methods with distinctive features for predicting EVCL. In experiments with a one-hour prediction scale, the best performance was obtained for EVCL prediction based on gated recurrent units (GRUs) [36]. Zhang Xian extracts the data features in the traffic flow by convolutional neural network, and then converts the data of EV arrival rate into EVCL by using a queuing model based on probabilistic ideas. The final results show that the model has high practical value and shows great potential for future practical applications [37].

C. Generative Adversarial Network

GAN can continuously output EVCL data with distribution patterns through the structure of adversarial. Wang Gan uses generative adversarial network (GAN) to process EVCL data, after which the random forest (RF) algorithm is used to extract the temporal load prediction data and fill the prediction data into the spatio-temporal matrix to complete the prediction of EVCL. GAN-RF is used in the simulation experiments with SVR and BP neural network models Stronger generalization capability was obtained in simulation experiments with SVR and BP neural network models [38]. Deng Jizhou constructed a hybrid EVCL prediction model using bi-directional generative adversarial network (BiGAN) and migration learning. The model has higher accuracy compared to the conventional model, while the model does not depend on the number of training data samples and can predict the load of charging stations for a small or even zero model samples [39]. Jaeuk Moon used conditional table GAN (CTGAN) to generate virtual calendar and temperature data, and later mixed the EVCL data generated based on deep learning to train another regression model to construct the EVCL prediction model [40]. Nadjib Mohamed constructed a conditional GAN model using temperature, date, and month data and obtained the smallest mean absolute percentage error on year-scale EVCL prediction experiments compared to other types of GAN networks [41].

V. CONCLUSIONS

In this study, a review of the development process of different forecasting techniques in the field of EVCL forecasting and the improvement of forecasting techniques is launched to help scholars and experts in the field. Usually, the electrical load of charging stations is affected by different factors such as meteorological conditions, economic activities, national policies and human behavior. To better manage charging station systems, build smart cities, and reduce carbon emissions, accurate charging station power load forecasting is

necessary. At this stage of research, the load prediction of charging stations can be predicted by constructing traffic networks from the perspective of electric vehicles, or by extracting features from the performance of electric vehicle load demand in the data. In the past few years, with the development of machine learning type algorithms, the algorithmic direction in the field of EVCL forecasting has changed considerably from the most basic statistical or data-based forecasting methods to those based on artificial intelligence techniques. EVCL prediction algorithms based on various AI techniques have shown great leadership in both prediction accuracy and prediction speed. Techniques based on GAN or reinforcement learning for load forecasting also show novel perspectives for modeling load forecasting problems.

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